Airbus Product Standard

EP-Prepreg

Woven Carbon Fiber Fabrics preimpregnated with Epoxy Resin

Contents

1 Scope

This standard specifies the required characteristics of EP-prepreg made from carbon fibre fabric preimpregnated with epoxy or modified epoxy resin for structural applications.

2 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

ISO2768-1	General tolerances - Tolerances for linear and angular dimensions without individual tolerance indications
ISO2768-2	General tolerances for features without individual tolerance indication
EN2424	Aerospace series - Marking of aerospace products
ABS5034	Airbus Standard - PF-Prepeg; Carbon fiber fabric preimpregnated with phenolic-formaldehyde resin compound
ABS5047	Airbus Standard-PF-Prepreg; Glassfiber preimpregnated materials for aircraft parts (e.g. interior) subject to Fire, Smoke and Toxicity (FST) requirements thermosetting (e.g. PF)
AIMS05-01-000 Part 1	Airbus Material Specification - Carbon fibre reinforced epoxy prepreg/ General requirements - Technical specification
AIMS05-01-000 Part 3	Airbus Material Specification - Carbon fibre reinforced epoxy prepreg Fabric / 180°C curing class - Qualification program / Batch release testing - Technical specification
AIMS05-01-003	Airbus Material Specification - Carbon fibre reinforced epoxy prepreg Plain weave fabric / 180°C curing class - Standard modulus fibre class - Material specification
AIMS05-01-004	Airbus Material Specification - Carbon fibre reinforced epoxy prepreg Five harness satin fabric / 180°C curing class - Standard modulus fibre class - Material specification
AIMS05-01-006	Airbus Material Specification - Carbon fibre reinforced epoxy prepreg Twill 2/2, 285 g/m² / 180°C curing class - Standard modulus fibre class - Material specification
AIMS05-01-010	Airbus Material Specification - Carbon fibre reinforced epoxy prepreg Five harness satin fabric 370g/m² / 180°C curing class - Standard modulus fibre class - Material specification
AIMS05-10-016	Airbus Material Specification - Fibre reinforced (woven) thermosetting preimpregnated materials for aircraft parts (e.g. interior) subject to Fire, Smoke and Toxicity (FST) requirements – thermosetting systems – carbon (3K carbon fibre, plain weave 1/1, 193 g/m²) resin mass content 53%
DAN1147	EP-prepreg: Woven fabric of carbon filament yarn 8.3541.80-1 preimpregnated with epoxy resin - 180°C curing system - Material standard
DAN1208–1	EP-prepreg: Woven fabric of carbon filament yarn preimpregnated with epoxy resin (EP-resin) class A, prepreg fabric; 125°C curing system-Material standard
DAN1208–2	EP-prepreg: Woven fabric of carbon filament yarn preimpregnated with epoxy resin (EP-resin) class B, prepreg fabric; 125°C curing system-Material standard

DAN1295	EP-prepreg: Woven fabric of carbon filament yarn preimpregnated with epoxy resin (EP-resin) prepreg fabric - 180°C curing system - Material standard
DAN432	Preimpregnated carbon fibre with epoxy resin, UD-prepreg and preimpregnated fabric -Technical specification
DAN480	Carbon fibre fabric preimpregnated with mod. epoxy resin co-curing system - Technical specification
WL8.3515.80	Aerospace; Carbon fibers; Woven fabric of carbon filament yarn, plain weave 1/1, 193 g/m^2
WL8.3540.80	Aerospace; Carbon fibers; Woven fabric of carbon filament yarn, satin weave 1/7, 370 g/m^2
WL8.3541.80	Aerospace; Carbon fibers; Woven fabric of carbon filament yarn, satin weave 1/4, 370 g/m^2

3 Requirements

3.1 Configuration, Dimensions, Tolerances and Mass

Dimensions and tolerances shall be in accordance to Table 1.

Length, roll inner diameter and mass per roll could be found in the general Technical Specification or in the Individual Product Specification (IPS) if necessary.

Table 1: Dimensions and tolerances

Material	Size Code	Width - trimmed [mm]
See Table 2, 3 and 4	0000	Width is not defined, width as agreed with the manufacturer
	1000	1000 ± 15
	1070	1070 ± 15
	1170	1170 ± 15
	1200	1200 ± 15
	1270	1270 ± 15
	1500	1500 ± 15

3.2 Material

See Tables 2, 3 and 4.

Table 2: EP-Prepreg, 180°C cure

Material code	Material	Cured ply thickness 1) [mm]	Mass per unit area [g/m^2]	Type of weave	Temperature range [°C]
A	AIMS05-01-003	0,205	325 ± 25	Plain weave 1/1	See AIMS
B	AIMS05-01-004	0,28	445 ± 25	5 harness satin fabric	
C	AIMS05-01-006A	0,31	490 ± 30	Twill 2/2	
D	AIMS05-01-006B		475 ± 25		
E	AIMS05-01-010A DAN1147-01	0,35	570 ± 35	WL8.3541.80-1 satin weave 1/4	See AIMS / DAN
F	AIMS05-01-010B DAN1147-02		620 ± 40		
G	DAN1295-01	0,235	345 ± 20	WL8.3515.80-1 plain weave 1/1	

Remarks:

1) Nominal value

Table 3: Prepreg, 135°C for co-curing 1)

Material code	Material	Cured ply thickness 2) [mm]	Mass per unit area [g/m^2]	Type of weave	Temperature range [°C]
H	AIMS05-10-016	0,28	411 ± 27	WL8.3515.80-1 plain weave 1/1	See AIMS

Remarks:

1) Application for co-curing with PF-prepreg per ABS5034, ABS5047-40 and ABS5047-51

2) Nominal value

Table 4: EP-Prepreg, 125°C cure

Material code	Material	Cured ply thickness [1] [mm]	Mass per unit area [g/m^2]	Type of weave	Temperature range [°C]
K	DAN1208A35	0,35	540 ± 30	WL8.3540.80-1 satin weave 1/7 or WL8.3541.80-1 satin weave 1/4	See DAN
M	DAN1208B35				
N	DAN1208A40		620 ± 40		
P	DAN1208B40				

Remarks:
1) Nominal value

4 Designation

This type of standard shall be designated according to the philosophy of the following example:

	Description block	Identity block			
	Prepreg-EP, CFRP-Fabric	ABS5003	A	0000	-03
Number of this standard		ABS5003			
Material code (See Table 2, 3, 4)			A		
Size code (See Table 1),				0000	
Blank when no IPS code required or IPS code* in accordance with IPS number (2 last digits) linked to AIMS material code					-03

* Note: designation with IPS code when required by the design

Table 5: Codes equivalence (Previous codes)

Column 1 will be applicable designation for drawing. In case that different drawing designation has been already used, column block 2 provide information of cross references applicable				
Column 1	**Column block 2** (previous designation)			
Material identity block	**Airbus France Code**	**Airbus Deutschland Code**	**Airbus UK Code**	**Airbus Spain Code**
ABS5003A0000-03				Z19.760 ABS5003M44EP205
ABS5003B0000		ABS5003N37EP280	ABS5003N40EP280	Z19.732 ABS5003N40EP280
ABS5003C0000		ABS5003R42EP250		Z19.733
ABS5003C0000-02	ABS5003R42-02			
ABS5003D0000-03	ABS5003R42-03			
ABS5003E0000		ABS5003F35EP350		
ABS5003F0000		ABS5003F40EP350		
ABS5003G0000		ABS5003E44EP250 ABS5003DAN1295-01EP250		
ABS5003H0000		ABS5003H53EP280 ABS5003DAN1180-02EP280		
ABS5003K0000		ABS5003A35EP350 ABS5003DAN1208A35EP350		
ABS5003M0000		ABS5003B35EP350 ABS5003DAN1208B35EP350		
ABS5003N0000		ABS5003A40EP350 ABS5003DAN1208A40EP350		
ABS5003P0000		ABS5003B40EP350 ABS5003DAN1208B40EP350		

5 Technical Specification

AIMS05-01-000 Part 1 and Part 3/ DAN432 / DAN480 as applicable.

Record of Revisions

Issue	Clause modified	Description of Modification
First publication 11.1984		New Standard
1 05.1994		Code E added
2 03.1998		Code F and H added
3 06.2000		Code M and N added
4 12.2004		Complete revision with new allocation of letters for material code, and addition of the table of previous codes (table 5)
5 02.2007	Table 3	DAN1180 has been changed to AIMS 05-10-016
6 03.2011	 2 3.1, Table 1	Corrected minor mistakes from issue 5 Added AIMS05-01-010 to DAN1147 1500mm width added, Table 1 moved to Chapter 3.1 from Chapter 3.2
7 04.2012	Table 1 Table 2	Width code 1170 and 1200 added Cured ply thickness of code C and D changed to 0,31 mm
8 07.2014	Table 2	Corrected minor typing mistakes
9 06.2018	All Table 1 Table 2	Reformatted to new template Size code 1070 added Mass per unit area (g/m^2) of the prepreg has been changed for the material code D (AIMS05-01-006B): $(470 \pm 30)\ g/m^2$ to $(475 \pm 25)\ g/m^2$